



Unit 3 -The Atom: Essential Knowledge and Guided Notes

Read Textbook: CK Chemistry Chapter 4: Atomic Structure
<https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

Watch Helpful Videos and Practice: Posted on CANVAS

Highlight Essential Knowledge: Use a different color marker or highlighter.

Atomic Structure Self Correcting Quizzes: <http://tinyurl.com/geekatomicstructure>
<http://tinyurl.com/geekatomicnumber> <http://tinyurl.com/geekatomicnumbers2>

- Fundamental particles that make up the atom include protons, neutrons, and electrons.
 - Protons (p^{1+})** have a positive (+) charge. They have a mass of approximately 1 amu. They are located in the nucleus of an atom.
 - Neutrons (n^0)** have no charge. They have a mass of approximately 1 amu. They are located in the nucleus of an atom.
 - Electrons (e^{1-})** have a negative (-) charge. They have a mass of approximately 0 amu (insignificant compared to the whole atom). They are located in electron clouds outside the nucleus.
- Protons and neutrons** are located in the nucleus of the atom and **comprise most of an atom's mass**.
- Quarks are also located in the nucleus of the atom and make up protons and neutrons.
- Atomic Number and Mass Number** are important terms to know.
 - The **atomic number** of an element is the same as the number of protons. In a neutral atom, the number of electrons is the same as that of protons. All atoms of an element have the same number of protons.
 - The **mass number** of an atom is the sum of the number of protons and neutrons. The number of neutrons in an atom can be found given its mass number.

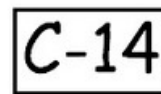
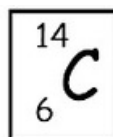
$$\# \text{ Neutrons} = \text{Mass Number} - \text{Atomic Number} = (p+n) - (p) = n$$
- An **isotope** is an atom with the same number of protons as another atom of the same element but a different number of neutrons. Some isotopes are radioactive; many are not. It is different for each element's isotopes. Each isotope has a different mass number.

Isotope Symbols

Mass Number → **A**
 Atomic Number → **Z**

$(\# \text{ of protons} + \# \text{ of neutrons})$
X
 $(\# \text{ of protons})$

← Element



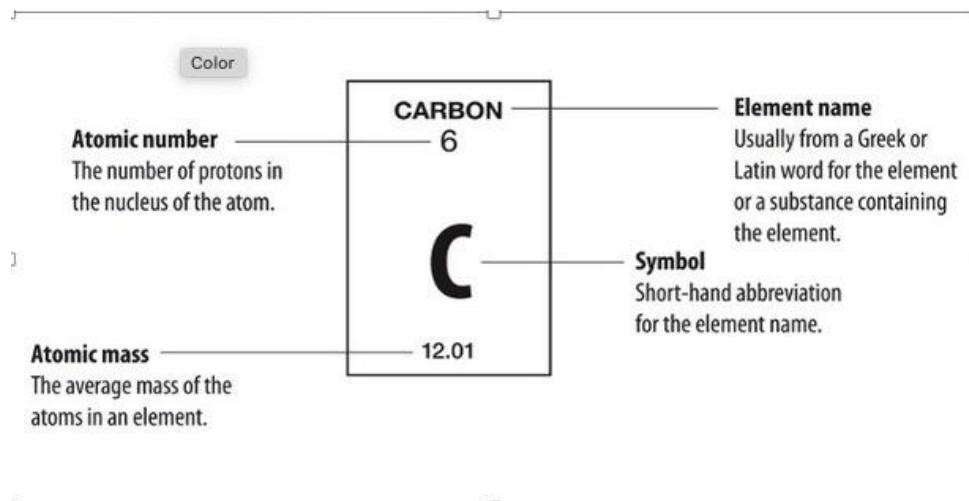
These both represent Carbon with 6 protons and 8 neutrons

The isotope symbol shows important information.

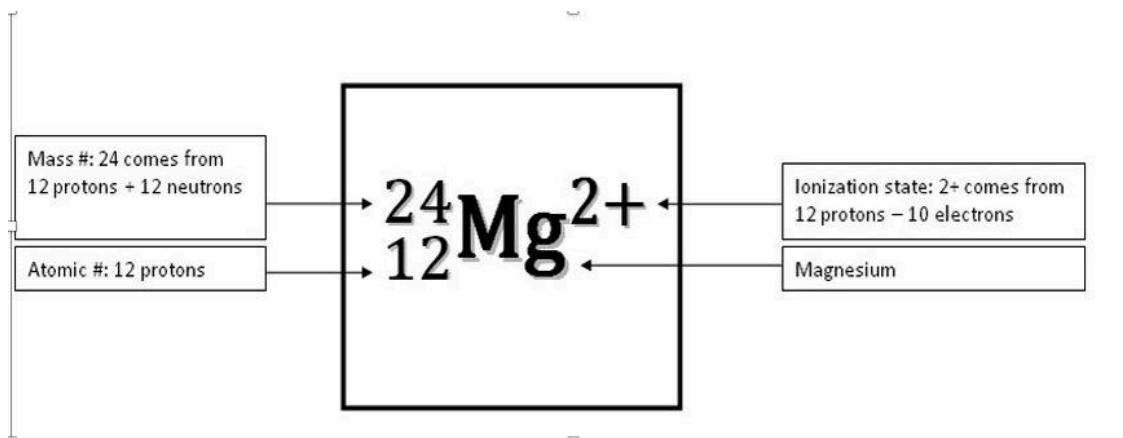
6. The relative abundance of an isotope is **the percentage of atoms with a specific atomic mass found in a naturally occurring sample of an element.**
7. The average **atomic mass** for each element is the weighted average of that element's naturally occurring isotopes. The isotopic abundance is used to calculate the atomic mass of an element. The formula for calculating atomic mass is shown below:

$$\text{Atomic mass} = \frac{(\% \text{ Isotope 1}) (\text{Atomic Mass Isotope 1}) + (\% \text{ Isotope 2}) (\text{Atomic Mass Isotope 2}) + \dots}{100}$$

8. The Periodic Table can be used to determine the atomic number, atomic mass, number of protons, the number of electrons, and the number of neutrons of any neutral atom of a particular element. The periodic table is arranged in order of increasing atomic numbers



9. An **ion** is an atom or a group of atoms that carry a positive or negative charge.
- The **charge** is sometimes called the **Ionization State**.
 - In a **neutral atom**, the number of electrons = number of protons
 - In a **positive ion** or **cation**, the number of electrons < number of protons
 - In a **negative ion** or **anion**, the number of electrons > number of protons
10. An ion symbol differs from an atom symbol in that it shows the charge or ionization state in the upper right corner:



Guided Notes: Lesson #1 - Atomic Anatomy

1.

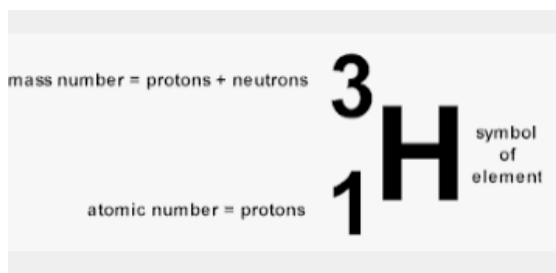
Subatomic Particle	Mass (AMU)	Charge	Location

2. What is an AMU?

3.

Term	Definition
Atomic Number	
Mass Number	
_____	All Atoms of the same element have the same number of protons. Isotopes are atoms of the same element that have different numbers of neutrons and different masses.

4. Notation:



5. Two ways of writing the _____ of Carbon
 Symbol-Mass Number C-12 or C-14

Mass Number Symbol ^{12}C or ^{14}C

6. The formula for determining the number of neutrons:

7. Can you figure out the number of neutrons and protons in Isotopes of Uranium? (Atomic #92)

Symbol	Mass Number	Protons	Neutrons
U-238			
U-235			

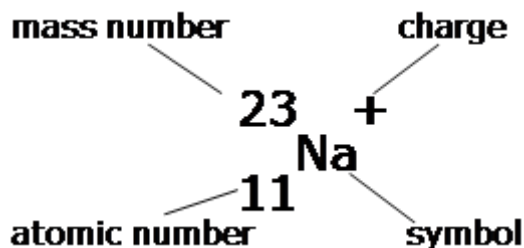
8. What is an ion?

Particle	Definition
Atom	Neutral, protons (+) ____ electrons (-)
Ion	_____ atoms
Cation	Cation (____) = Protons____Electrons
Anion	Anion (____) = Protons____Electrons

9. **Practice**
Charge = protons – electrons

1. A sodium ion has 11 protons and 10 electrons.
 - a. What is its charge?
 - b. Is it a cation or an anion?
2. A chloride ion has 17 protons and 18 electrons.
 - a. What is its charge?
 - b. Is it a cation or an anion?

10. Notation for ions:



11. Fill in the table to practice

Symbol	Atomic Number (p)	Mass Number (p + n)	Charge (p-e)	# protons	# neutrons	# electrons	Atom, Cation Or Anion?
^7Li							
	11	23	+1	11	12	10	cation
$^{32}\text{S}^{2-}$							
	17	37	0	17	20	17	atom

12. What is a quark?

Guided Notes: Lesson #2 - Atomic Mass

1. Determine your test average from 5 tests: 100, 100, 100, 100, 50
2. Here are 4 methods to do it. Do they all work?

METHOD 1

$$\frac{100 + 100 + 100 + 100 + 50}{5} =$$

METHOD 2

$$\frac{4(100) + 1(50)}{5} =$$

METHOD 3

$$\frac{80(100) + 20(50)}{100} =$$

METHOD 4

$$\text{Or } .80(100) + .20(50) =$$

3. Try this problem: What is your average for the quarter? _____

% of Grade	Average
40% Assignments	85
60% Tests	95

4. What are the 3 main isotopes of carbon? Fill in the table below

Isotope	Protons	Neutrons	Electrons
C-12			
C-13			
C-14			

What is its atomic mass? _____

5. Fill in the table

Term	Definition
	# of protons
	number of protons + neutrons in a single isotope
	the weighted average of all the isotope masses. On the periodic table, it is the mass in grams of 1 mole (6.0×10^{23} atoms) of the substance

6. Can you find the atomic mass of Lithium?

$$\text{Atomic Mass} = \text{Weighted Average} = \frac{(\%_1)(\text{Mass}_1) + (\%_2)(\text{Mass}_2)}{100}$$

Isotope Symbol	Mass of Isotope (AMU)	Relative Abundance (%)
Li-6	6.015	7.49
Li-7	7.016	92.51

7. Is there any atom of lithium that has a mass of 6.94 amu?

8. How do you find the mass of an atom?

9. Do Cl-35 or Cl-37 have the same atomic mass as Cl on the periodic table? Why or why not?